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Article in *The Journal of Strength and Conditioning Research* · July 2023

DOI: 10.1519/JSC.0000000000004404

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AU2

Acute Effects of Isometric Conditioning Activity on the Viscoelastic Properties of Muscles and Sprint and Jumping Performance in Handball Players

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Abstract

Krzysztofik, M., Spieszny, M., Trybulski, R., Wilk, M., Pisz, A., Kolinger, D., Filip-Stachnik, A., and Stastny, P. Acute effects of isometric conditioning activity on the viscoelastic properties of muscles and sprint and jumping performance in handball players. *J Strength Cond Res* XX(X): 000–000, 2022—The effects of conditioning activity (CA) on muscle stiffness are currently unknown, suggesting that maximum CA effort can increase or decrease the stiffness of involved muscle groups. Therefore, this study aimed to investigate the effect of maximal isometric half-squats on the viscoelastic properties of muscles and postactivation performance enhancement (PAPE) in sprints and jumps. Twelve handball players underwent a standard warm-up and baseline assessment of muscle stiffness and tone of vastus lateralis and gastrocnemius medialis muscle, followed by 20-m sprint with intermediate measures at 5 and 10 m and countermovement jump. The PAPE was assessed by repeating the tests (at 4th, 8th, and 12th minute post-CA) after a CA protocol consisting of 3 sets of 3-second maximal isometric half-squats (EXP) or a control condition (CTRL) without any CA. The vastus lateralis stiffness in the 4th and 12th minute and muscle tone in the 4th minute post-CA significantly decreased compared with baseline ($p = 0.041$, $ES = 0.57$; $p = 0.013$, $ES = 0.52$; $p = 0.004$, $ES = 0.81$, respectively) in the EXP condition. The 20-m sprint time significantly decreased at all post-CA time points compared with the baseline for the EXP condition ($p < 0.033$) and the after values in the CTRL condition ($p < 0.036$). In comparison to baseline, the 10-m sprint time decreased in the eighth minute post-CA ($p = 0.021$; $ES = 0.82$) in the EXP condition. Moreover, it was significantly lower at the 8th and 12th minute post-CA ($p = 0.038$; $ES = 0.71$ and $p = 0.005$; $ES = 1.26$) compared with that time points in the CTRL condition. The maximal isometric half-squats effectively improved sprint performance and significantly decreased vastus lateralis tone and stiffness. These findings offer new insights into the assessment of viscoelastic properties for evaluating the fatigue or potentiation state, which requires further investigation.

Key Words: resistance training, PAP, PAPE

AU3

AU4

Introduction

Research on postactivation performance enhancement (PAPE) has attracted interest from the scientific community and sports practitioners for a decade due to the unique possibility of acute improvement in performance due to PAPE. Often, this is achieved through maximum voluntary contractions as a condition activity (CA) with similar biomechanical demands as a subsequent explosive task, e.g., performing a back squat exercise before vertical jumping (42). Usually, studies are focused on examining the effectiveness of a wide variety of CAs for contraction types, intensities, and volumes, along with different rest intervals (8,29,36,42,44), in inducing improvements in subsequent activities. Moreover, it seems that the magnitude of the PAPE effect may differ depending on the primary task because the literature reveals that the greatest increases in performance are noted in sprints and somewhat less in jumping tasks (42).

Recent studies have focused primarily on the assessment of changes in contractile properties of muscles due to the applied CA (6,22,37). However, as a result of PAPE protocols, muscles' viscoelastic properties may also be acutely affected (18). Despite previous speculations about the possible impact of stiffness on the PAPE effect (5,7), only 2 studies have focused on assessing the relationships between those phenomena (18,38). A study by Gago et al. (18) showed no change in Achilles tendon stiffness despite the PAPE effect, whereas Pożarowski et al. (38) reported increased stiffness after a CA. The authors did not evaluate subsequent athletic performance; thus, this issue remains unresolved.

In turn, much more research has been performed on the effect of fatigue on viscoelastic properties of muscles and tendons (23,27,28,45,46). Most of these studies are consistent and indicate an increase in stiffness due to fatigue (27,28,45,46). Furthermore, it has been suggested that assessing subtle changes in muscle stiffness may help identify early development of muscle fatigue (2,43). Because the stretch-shortening cycle is strongly related to stiffness (30), from the theoretical point of view, an increase in stiffness should positively affect physical fitness. However, the opposite was shown in the studies mentioned

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Journal of Strength and Conditioning Research 00(00)/1–9

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above; a decrease in performance was accompanied by increased stiffness. It is not entirely clear what mechanisms underlie this phenomenon. It is speculated that this may be a compensatory or protective mechanism for the body against fatigue. Increased stiffness may be accompanied by microswelling within the muscle fibers, which was reported by Proske and Morgan (39). Another explanation may be a fatigue-induced increase in muscle metabolites causing myosin cross-bridges to adhere more tightly to the titin fibers (13,48) and a reduced relaxation time, thus a lower cross-bridge detachment rate (1,16,47).

The current research provides contradictory results for muscle stiffness changes after the PAPE effect; the results are also contradictory as to whether PAPE or fatigue is related to the increase or decrease in muscle stiffness. Therefore, this study aimed to investigate the effects of maximal isometric half-squats on viscoelastic muscle properties (muscle tone and stiffness, as a primary outcome) and whether this CA would enhance subsequent short sprint (5, 10, and 20 m) and countermovement jump (CMJ) performance in handball players. We hypothesized that the maximum isometric back squat as a CA would increase the stiffness of the vastus lateralis (VL) and gastrocnemius medialis (GM) along with acute enhancement of 20-m sprint and jump performance. Bearing in mind the coexistence of PAPE and fatigue, it can be assumed that detecting a time point at which fatigue has subsided will increase the chances of performance improvement as long as the potentiated state occurs.

AU5 Methods

AU6 Experimental Approach to the Problem

The experiment was performed following a randomized crossover design where maximal isometric half-squats as CA were the experimental condition (EXP) and no CA was the control condition (CTRL). This setup aimed to investigate the effects of maximal isometric half-squats on viscoelastic muscle properties (muscle tone and stiffness) and subsequent CMJ and 20-m sprint performance among handball players. Therefore, muscle tone and stiffness were always measured as the first parameter (Figure 1). The randomization was performed for 2 different conditions using automatic software (www.randomizer.org) (Figure 1).

The VL and GM muscle viscoelastic properties, consequent 5-, 10-, and 20-m sprint, and CMJ performance were assessed 5 minutes before (pre-CA) and at the 4th, 8th, and 12th minute after the completion of the maximum isometric half-squat (post-CA) for EXP. In the CTRL, the same measurements were performed, but no CA was applied. In addition to the post-CA time points listed, the best post-CA performance was also retained for further studies to account for the large interindividual variability (12,14,20) in PAPE responses.

AU7 Subjects

Fifteen male handball players were selected to participate in this study. The inclusion criteria were as follows: (a) positive response to the used CA, (b) free from neuromuscular disorders, (c) no lower-limb surgery for the past 3 years, (d) have at least 2 years' experience in resistance training, (e) take part in regular handball training and competition for the last year, and (f) consider themselves safe to participate in the exercise protocol included in the study design as determined by the self-administered Physical Activity Readiness Questionnaire. Twelve subjects met the inclusion criteria (age: 20.8 ± 2.2 years, body mass: 85.3 ± 11 kg, body height: 181 ± 5 cm,

handball training experience: 11.1 ± 2.2 years, resistance training experience: 5 ± 2.3 years, relative back squat strength: 1.73 ± 0.21 kg·bm⁻¹). Subjects were instructed not to perform any resistance exercises 72 hours before testing to avoid fatigue, maintain their usual dietary and sleep habits, and not use any stimulants during the study. Subjects were allowed to withdraw from the experiment at any time and were informed of the benefits and potential risks of the study before providing their written informed consent for participation. Subjects were not told of the expected study outcomes. The study was conducted in accordance with the Declaration of Helsinki and approved by the Institutional Review Board of the Academy of Physical Education in Katowice (January 2021). The sample size was determined using G*Power version 3.1.9.2 (Dusseldorf, Germany), and the following parameters "ANOVA, repeated measures, and within factors" were assumed as a statistical test (1 group of subjects, 2 experimental conditions, and 4 measurements); the statistical power was 0.8, the significance level was 0.05, and the effect size was 0.51–1.13 based on previous studies that investigated acute changes in muscle stiffness after exercise (37) and the PAPE effect on sprinting performance (8). The power analysis indicated that a minimum sample size of between 4 and 8 individuals was required for this study.

Experimental Sessions. The subjects performed 2 different testing conditions (1 week apart) in a randomized order. The same typical pretraining warm-up was performed before each experimental session, which consisted of 5 minutes of jogging (forward, backward, and sideways) followed by 5 minutes of passing and throwing the ball into the goal. This was followed by 2 submaximal attempts of CMJ without arm swing, 20-m sprints, and 3-second isometric back squats. After that, the tests were performed in the following order: baseline viscoelastic muscle properties and performance measurements, CA, post-CA evaluation of viscoelastic muscle properties, and performance assessment at the 4th, 8th, and 12th minute after the CA. Previous studies used similar post-CA rest intervals (9,26,32). The same measurements were made at the same time points under the CTRL condition, but no CA was used (pre-CA and post-CA assessments of viscoelastic muscle properties and CMJ and 20-m sprint performance).

Conditioning Activity. After a standardized warm-up and baseline assessments, the subjects performed the barbell half-squat with a 90° knee angle as the CA. Before the testing, the individual height of the racks that were needed to obtain a 90° knee flexion joint was determined (by a goniometer) for each subject. The maximum isometric squat was performed on an unmovable barbell loaded with a supramaximal load (150% 1RM) that did not allow any concentric movement. Subjects went under the barbell and were instructed to push it as forcefully and fast as possible for 3 seconds. Three sets of 3 maximal attempts were performed with a 3-minute rest in between sets. This CA has been previously indicated to be effective in inducing an improvement in physical performance (11).

Measurement of Viscoelastic Muscle Properties. The handheld myometer (MyotonPRO, Myoton AS, Tallinn, Estonia) was used for the noninvasive assessment of viscoelastic muscle properties of VL and GM of the dominant limb through superficial mechanical deformation. The device's probe was placed perpendicular to the skin and brief mechanical compression was applied (0.4 N for 15 ms), with a constant preload force of 0.18 N. After a mechanical impulse, the soft tissue responds in the form of a dampened oscillation, which is measured by an accelerometer. The

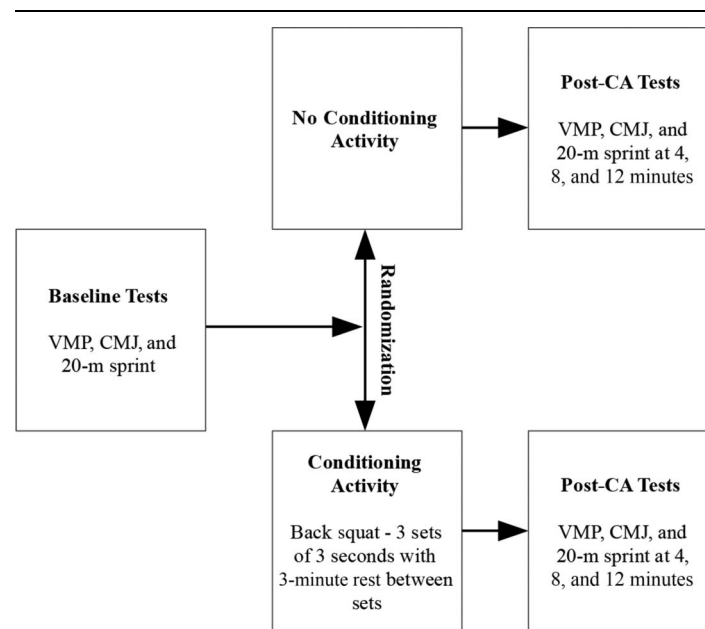


Figure 1. Study design. VMPs = viscoelastic muscle properties; CMJ = counter-movement jump; CA = conditioning activity.

measurement for VL was at 50% of the straight-line distance between the greater trochanter and fibulae capitulum (10), whereas for GM, it was 70% of the lower leg length measured from the lateral malleolus to the popliteal fossa (25). Measurements were made in a state of muscle relaxation with subjects lying in a prone position for the GM and lying in a supine position for the VL with their arms resting beside the body. The following viscoelastic muscle properties were evaluated: muscle tone (oscillation frequency) and stiffness.

Measurement of Sprinting Performance. Sprint times were recorded using timing photocells (SmartSpeed Pro, Fusion Sport, Coopers Plains, Australia), with gates at 0, 5, 10, and 20 m. The height was set at approximately 1 m off the ground, corresponding to subjects' hip height to avoid the timing gates being triggered prematurely by a swinging arm or leg. The subjects started with a front foot placed 0.3 m behind the first timing gate to prevent any early triggering of the photocells. Times were measured to the nearest 0.001 seconds. Two attempts were performed, and the best 20-m sprint performance was kept for further analysis.

Measurement of Jumping Performance. Jumping assessments were performed on a force plate (ForceDecks, Vald Performance, Australia) with a sampling rate of 1,000 Hz, which has been previously confirmed as valid and reliable (21). Each subject performed 2 CMJs without an arm swing at each time point. The subject started in the standing position with hands placed on the hips for this measurement. Then, they dropped into the counter-movement position to a self-selected depth and immediately followed by a maximal effort vertical jump. The subjects were instructed to land in the same position as the take-off in the midsection of the force plate. The subjects reset to the starting position after each jump, and the procedure was repeated for a total of 2 jumps. The jump height from flight time ($9.81 \times [\text{flight time}]^2/8$) and peak velocity were calculated. The best jump after post-CA for height was kept for further analysis.

Statistical Analysis

A Shapiro-Wilk test was used to test the normal distribution of the data, and Mauchly's test was used to test for the assumption of sphericity. A single rater intraclass correlation coefficient (ICC) and coefficient of variability (CV) were used to measure value reliability.

Two-way ANOVAs ($2 \times [\text{EXP}; \text{CTRL}] \times 4$ time points [pre-CA; 4, 8, and 12 minutes post-CA]) were used to investigate the influence of CA on viscoelastic muscle properties and sprinting and jumping performance. An additional 2-way ANOVA ($2 \times [\text{EXP}; \text{CTRL}] \times 2$ time points [pre-CA; best post-CA]) was used to examine individual peak PAPE responses. When a significant main effect or interaction was found, post hoc tests with Bonferroni correction were used to analyze the pairwise comparisons. The magnitude of mean differences was expressed with standardized effect sizes. Thresholds for qualitative descriptors of Hedge's g were interpreted as ≤ 0.20 "small," $0.21\text{--}0.79$ "medium," and > 0.80 as "large." The Pearson correlation coefficient was used to analyze the relationship between changes in athletic performance and viscoelastic muscle properties and between the magnitude of PAPE response and baseline viscoelastic muscle properties. Thresholds for qualitative descriptors of correlations were interpreted as follows: < 0.1 "trivial," $0.1\text{--}0.3$ "small," $< 0.31\text{--}0.5$ "moderate," $< 0.51\text{--}0.7$ "large," $< 0.71\text{--}0.9$ "very large," and > 0.9 "nearly perfect."

All data were analyzed using SPSS (version 25.0; SPSS, Inc., Chicago, IL) and are reported as the mean with standard deviations and their 95% confidence intervals (CIs). Statistical significance was set at $p < 0.05$.

Results

The Shapiro-Wilk tests did not indicate any violation of the data distribution of jumping and sprinting performance, muscle tone, or stiffness. The measurement reliability was high for muscle tone (ICC = 0.88, CV = 2.7%), muscle stiffness (ICC = 0.91, CV = 2.6%), sprinting performance (ICC = 0.82, CV = 1.6%), and

jumping performance (ICC = 0.89, CV = 2.9%). The best 20-m sprint performance was observed in the eighth $\pm$ 3:02 minutes after the CA, whereas for the CMJ, it was in the seventh $\pm$ 2:57 minutes.

Viscoelastic Muscle Properties

Vastus Lateralis. The 2-way ANOVA indicated that there was no statistically significant interaction of stiffness ($F = 0.441$; $p = 0.725$; $\eta^2 = 0.039$) and muscle tone ($F = 0.738$; $p = 0.537$; $\eta^2 = 0.063$). There was no main effect of condition on stiffness ($F = 0.108$; $p = 0.748$; $\eta^2 = 0.01$) or muscle tone ($F = 0.608$; $p = 0.452$; $\eta^2 = 0.052$). However, there was a main effect of time on stiffness ($F = 5.722$; $p = 0.003$; $\eta^2 = 0.342$) and muscle tone ($F = 4.571$; $p = 0.009$; $\eta^2 = 0.294$).

The post hoc analysis showed a significant decrease in muscle stiffness in the 4th and 12th minute post-CA compared with pre-CA ($p = 0.041$, ES = 0.57 and $p = 0.013$, ES = 0.52, respectively) in the EXP condition. The post hoc analysis showed a significant decrease in muscle tone in the fourth minute post-CA compared with pre-CA ($p = 0.004$, ES = 0.81) in the EXP condition (Table 1).

T1

Peak Performance. The 2-way ANOVA for peak PAPE response showed a statistically significant interaction for muscle tone ($F = 5.188$, $p = 0.044$, $\eta^2 = 0.32$) but not for stiffness ($F = 1.572$, $p = 0.236$, $\eta^2 = 0.125$), yet there was a main effect of time ($F = 10.273$, $p = 0.008$, $\eta^2 = 0.483$). The post hoc analysis revealed a significant decrease in muscle tone and stiffness post-CA compared with pre-CA ($p = 0.001$, ES = 0.75, and $p = 0.003$, ES = 0.54, respectively) in the EXP condition (Figure 2).

F2

Gastrocnemius Medialis. No statistically significant interaction for muscle stiffness ($p = 0.717$; $F = 0.453$; $\eta^2 = 0.04$) or tone ($p = 0.705$; $F = 0.470$; $\eta^2 = 0.041$) was found. Moreover, there were no statistically significant main effects of condition and time on muscle stiffness ($p = 0.361$; $F = 0.907$; $\eta^2 = 0.076$ and $p = 0.108$; $F = 2.189$; $\eta^2 = 0.166$, respectively) or muscle tone ($p = 0.797$; $F = 0.069$; $\eta^2 = 0.006$ and $p = 0.511$; $F = 0.785$; $\eta^2 = 0.067$, respectively) (Table 1).

Peak Performance. No statistically significant interaction for muscle stiffness ($p = 0.585$; $F = 0.316$; $\eta^2 = 0.028$) or tone ($p = 0.464$; $F = 0.575$; $\eta^2 = 0.05$) was found. Moreover, there

were no statistically significant main effects of condition and time on muscle stiffness ($p = 0.395$; $F = 0.782$; $\eta^2 = 0.066$ and $p = 0.322$; $F = 1.075$; $\eta^2 = 0.089$, respectively) or muscle tone ($p = 0.817$; $F = 0.056$; $\eta^2 = 0.005$ and $p = 0.5$; $F = 0.486$; $\eta^2 = 0.042$, respectively) (Figure 3).

F3

Sprinting Performance

A repeated measures 2-way ANOVA determined a statistically significant interaction for 10 m ($p = 0.029$; $F = 3.39$; $\eta^2 = 0.236$) and 20 m ($p = 0.003$; $F = 5.572$; $\eta^2 = 0.336$) but not for 5-m sprint time ($p = 0.127$; $F = 2.044$; $\eta^2 = 0.157$). However, the main effect of the condition for sprint performance at 5 m ($p = 0.025$; $F = 6.782$; $\eta^2 = 0.381$) was found.

Post hoc analysis revealed a statistically significant decrease in sprint time at 5 m at the 12th minute post-CA for the EXP condition ($p = 0.022$; ES = 0.93) compared with the CTRL condition. In addition, a statistically significant decrease in 10-m sprint time compared with pre-CA at the eighth minute was noted ($p = 0.021$; ES = 0.82) during the EXP condition. Moreover, there was a statistically significant decrease in 10-m sprint time during the EXP condition compared with the CTRL condition at the 8th and 12th minute post-CA ($p = 0.038$; ES = 0.71 and $p = 0.005$; ES = 1.26). The 20-m sprint time decreased significantly at all post-CA time points ($p < 0.033$; ES = from 0.66 to 0.85) compared with pre-CA for the EXP condition. In addition, it significantly decreased in comparison to all corresponding time points during the CTRL condition ($p < 0.036$; ES = from 0.8 to 1.4) (Table 2).

T2

Peak Performance. A repeated measures 2-way ANOVA revealed a statistically significant interaction for 10 m ($p = 0.018$; $F = 7.72$; $\eta^2 = 0.412$) and 20 m ($p = 0.001$; $F = 19.689$; $\eta^2 = 0.642$) but not for 5 m ($p = 0.065$; $F = 4.185$; $\eta^2 = 0.276$). However, there was a statistically significant main effect of time for a sprint at 5 m ($p = 0.009$; $F = 9.844$; $\eta^2 = 0.472$).

Post hoc analysis revealed that 5-, 10-, and 20-m sprint times significantly decreased from pre-CA to post-CA measures in the EXP condition ($p = 0.001$, ES = 1.42; $p = 0.001$, ES = 1.28; $p < 0.0001$, ES = 1.25, respectively) but not in the CTRL condition. Moreover, the post-CA sprint time at 10 and 20 m was

Table 1
Comparison of jumping and sprinting performance.*†

	Pre-CA		4 min post-CA		8 min post-CA		12 min post-CA	
	CTRL	EXP	CTRL	EXP	CTRL	EXP	CTRL	EXP
5 m (s)	1.051 $\pm$ 0.058 (1.014–1.088)	1.058 $\pm$ 0.037 (1.035–1.082)	1.05 $\pm$ 0.044 (1.021–1.078)	1.031 $\pm$ 0.044 (1.003–1.059)	1.047 $\pm$ 0.039 (1.022–1.072)	1.031 $\pm$ 0.042 (1.004–1.057)	1.051 $\pm$ 0.037 (1.027–1.074)	1.002 $\pm$ 0.062‡ (0.963–1.042)
10 m (s)	1.82 $\pm$ 0.078 (1.771–1.87)	1.823 $\pm$ 0.053 (1.789–1.857)	1.814 $\pm$ 0.054 (1.779–1.849)	1.781 $\pm$ 0.067 (1.738–1.823)	1.817 $\pm$ 0.055 (1.782–1.853)	1.773 $\pm$ 0.064‡§ (1.732–1.814)	1.835 $\pm$ 0.039 (1.81–1.859)	1.745 $\pm$ 0.089‡ (1.688–1.802)
20 m (s)	3.142 $\pm$ 0.102 (3.077–3.207)	3.136 $\pm$ 0.108 (3.067–3.205)	3.143 $\pm$ 0.08 (3.092–3.194)	3.058 $\pm$ 0.121‡§ (2.981–3.135)	3.137 $\pm$ 0.068 (3.094–3.181)	3.042 $\pm$ 0.106‡§ (2.974–3.109)	3.19 $\pm$ 0.063 (3.149–3.23)	3.049 $\pm$ 0.122‡§ (2.972–3.126)
Jump height (cm)	37.9 $\pm$ 3.2 (35.9–39.9)	38.4 $\pm$ 3.2 (36.4–40.5)	37.9 $\pm$ 2.8 (36.1–39.6)	38.7 $\pm$ 3.3 (36.6–40.8)	37.6 $\pm$ 2.8 (35.8–39.4)	38.8 $\pm$ 3.5 (36.5–41)	37.6 $\pm$ 2.7 (35.9–39.3)	39 $\pm$ 3.6 (36.7–41.3)
Peak velocity (m·s ⁻¹)	2.78 $\pm$ 0.12 (2.7–2.86)	2.81 $\pm$ 0.13 (2.72–2.89)	2.79 $\pm$ 0.13 (2.71–2.87)	2.79 $\pm$ 0.12 (2.71–2.86)	2.79 $\pm$ 0.13 (2.72–2.87)	2.83 $\pm$ 0.13 (2.76–2.91)	2.79 $\pm$ 0.11 (2.72–2.86)	2.81 $\pm$ 0.15 (2.72–2.91)

*CA = conditioning activity; CTRL = control condition; EXP = experimental condition.

†Values are presented as mean $\pm$ SD.

‡Significant difference in comparison to CTRL.

§Significant difference in comparison to pre-CA.

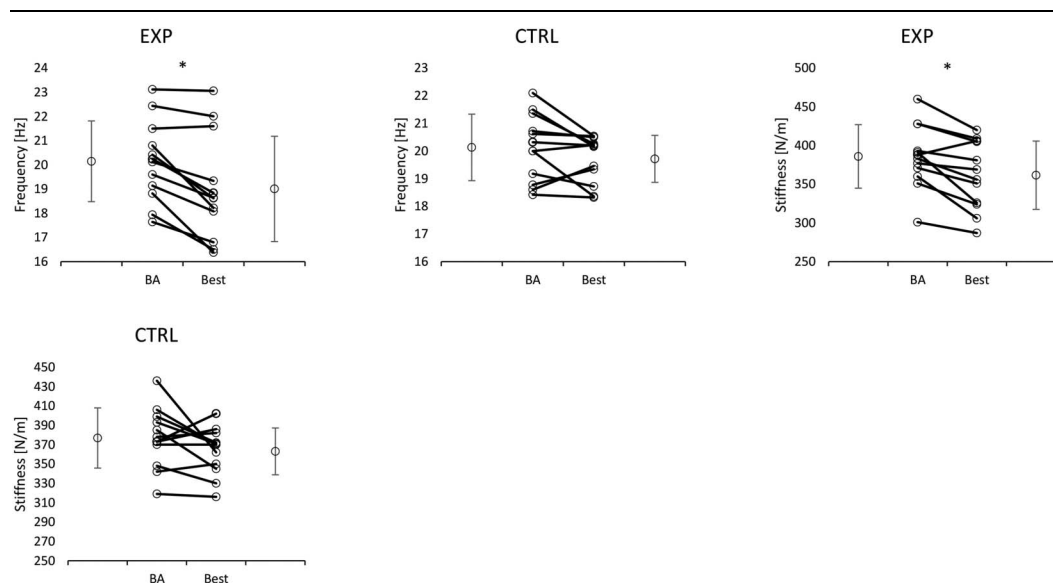


Figure 2. Mean $\pm$ SD and interindividual responses to the CA in viscoelastic properties of vastus lateralis. CA = conditioning activity; BA = baseline; BEST = best performance postconditioning activity irrespective of the rest interval; EXP = experimental condition; CTRL = control condition.

significantly lower in the EXP condition than in the CTRL condition ($p = 0.004$, ES = 0.97, and $p = 0.001$, ES = 1.32) (Figure 4).

F4

Jumping Performance

No statistically significant interaction for jump height ($p = 0.477$; $F = 0.849$; $\eta^2 = 0.072$) or peak velocity ($p = 0.869$; $F = 0.239$; $\eta^2 = 0.021$) was found. Moreover, there were no statistically significant main effects of condition and time on jump height ($p = 0.054$; $F = 4.645$; $\eta^2 = 0.297$ and $p = 0.96$; $F = 0.1$; $\eta^2 = 0.009$, respectively) or peak velocity ($p = 0.131$; $F = 2.666$; $\eta^2 = 0.195$ and $p = 0.733$; $F = 0.429$; $\eta^2 = 0.038$, respectively) (Table 2).

Peak Performance. A repeated measures 2-way ANOVA revealed a statistically significant interaction for jump height ($p = 0.002$; $F = 15.716$; $\eta^2 = 0.588$) but not for peak velocity ($p = 0.545$; $F = 0.391$; $\eta^2 = 0.034$). However, there was a statistically significant main effect of time ($p = 0.001$; $F = 19.839$; $\eta^2 = 0.643$) and condition ($p < 0.001$; $F = 24.378$; $\eta^2 = 0.689$) for peak velocity.

Post hoc analysis revealed that jump height significantly increased from pre-CA to post-CA in the EXP condition ($p < 0.0001$; ES = 0.5) but not in the CTRL condition. Moreover, post-CA jump height was significantly higher in the EXP condition than in the CTRL condition ($p = 0.004$; ES = 0.57). No statistically significant difference was found for peak velocity (Figure 5).

F5

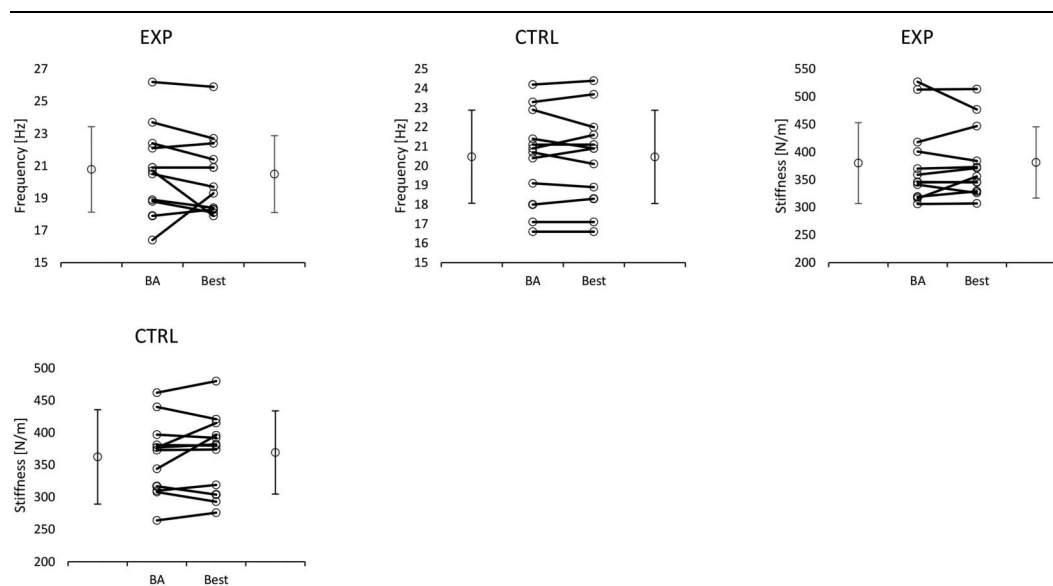


Figure 3. Mean $\pm$ SD and interindividual responses to the CA in viscoelastic properties of gastrocnemius medialis. CA = conditioning activity; BA = baseline; BEST = best performance postconditioning activity irrespective of the rest interval; EXP = experimental condition; CTRL = control condition.

Table 2**Comparison of pre-CA and post-CA viscoelastic muscle properties of the vastus lateralis muscle.*†**

	Pre-CA		4 min post-CA		8 min post-CA		12 min post-CA	
	CTRL	EXP	CTRL	EXP	CTRL	EXP	CTRL	EXP
Vastus lateralis								
Muscle tone (Hz)	20.13 ± 1.2 (19.37–20.9)	20.15 ± 1.67 (19.09–21.21)	19.64 ± 1.9 (18.43–20.84)	18.91 ± 2.12 (17.56–20.26)‡	19.58 ± 1.18 (18.83–20.32)	19.38 ± 2 (18.11–20.65)	19.86 ± 1.44 (18.95–20.77)	19.47 ± 2.68 (17.77–21.17)
Stiffness (N·m ⁻¹)	377 ± 31 (357–397)	386 ± 41 (360–412)	365 ± 39 (341–390)	361 ± 43 (334–389)‡	364 ± 25 (349–380)	366 ± 45 (338–395)	358 ± 33 (337–379)	361 ± 52 (328–394)‡
Gastrocnemius medialis								
Muscle tone (Hz)	20.48 ± 2.41 (18.95–22)	20.78 ± 2.65 (19.09–22.47)	20.48 ± 2.3 (19.01–21.94)	20.37 ± 2.63 (18.7–22.03)	20.59 ± 2.37 (19.09–22.1)	20.8 ± 2.34 (19.31–22.29)	20.28 ± 1.93 (19.06–21.51)	20.6 ± 2.48 (19.02–22.18)
Stiffness (N·m ⁻¹)	363 ± 57 (326–399)	380 ± 73 (334–427)	363 ± 53 (329–396)	384 ± 57 (336–431)	378 ± 68 (335–421)	386 ± 62 (346–425)	371 ± 63 (331–411)	387 ± 67 (344–429)

*CA = conditioning activity; CTRL = control condition; EXP = experimental condition.

†Values are presented as mean ± SD.

‡Significant difference in comparison to pre-CA.

Relationships Between Muscle Stiffness and Postactivation Performance Enhancement

The Pearson correlation coefficient indicated no statistically significant relationships between changes in viscoelastic muscle properties at baseline and the magnitude of the PAPE response.

Discussion

The primary findings of this study are that the maximal isometric half-squat CAs change the VL viscoelastic muscle properties with a concurrent improvement of 20-m sprint performance and no concurrent changes of GM viscoelastic muscle properties. Explicitly, the maximal isometric half-squat CAs contributed to the significant decrease in VL tone and stiffness at the fourth minute and VL stiffness at the 12th minute post-CA, with no effect on GM viscoelastic properties at any time point. In addition, the used CA led to a significant improvement of 20-m sprint time from 4th to 12th minute and in 10-m sprint time at 8th minute after CA compared with the CTRL condition; however, no CA effects in CMJ performance and 5-m sprint time were noted. This adds a novel perspective to muscle viscoelastic properties concerning PAPE because the results of previous studies have reported a substantial increase in muscle stiffness due to fatiguing protocols (27,28,45,46). On the other perspective, our result shows that a muscle tone and stiffness decrease might accompany muscle potentiation during the resting period, which requires further investigation to establish the mechanisms underlying this phenomenon.

Our secondary outcome, that isometric half-squat CAs increase the sprint performance, was used as the initial premise to expect the changes in muscle viscoelastic properties and to confirm that the CA protocol is more beneficial than CTRL. Although we have not found the effect in CMJ, and we did not confirm our hypothesis in muscle viscoelastic properties, it can be assumed that the level of fatigue caused by the applied CA was low. Suggesting that the mechanisms underlying the PAPE effect dominated and contributed to the acute improvements in sprint performance, considering that isometric contractions show a lower metabolic cost than dynamic contractions (40,41). Hence, isometric contractions may be more efficient in inducing the PAPE effect, leading to less metabolic disturbances and, consequently, less fatigue (11,40). This may be confirmed by the noted decrease

in muscle stiffness and muscle tone, considering that previous studies have found an increase in muscle stiffness due to strenuous protocols (27,28,45,46). Therefore, it seems that muscle stiffness and muscle tone changes (assessed through myotonometry) are primarily related to the recovery mechanisms rather than to the mechanisms eliciting the PAPE effect. This can also be partially confirmed by the results of the studies by Gago et al. (18,19), in which no changes in Achilles stiffness were recorded despite an increase in twitch force. Nevertheless, it may be indirectly helpful for verifying the effectiveness of the CA used to elicit the PAPE effect. In addition, although a significant decrease in viscoelastic VL properties was reported, it was not significantly correlated with the enhancement in athletic performance. Therefore, the viscoelastic properties of the muscles involved in the CA may indicate changes in performance but not reflect their magnitude. Perhaps the viscoelastic properties could help determine an individual's optimal rest interval between CA and the subsequent explosive task, as long as the appropriate level of potentiation is triggered. However, far more studies evaluating different PAPE protocols in different populations are necessary to draw definitive conclusions.

As a part of the primary outcome, there were notably no significant changes in the stiffness and tone of the GM muscle, although they were present in the VL muscle. The ankle muscles are responsible for creating a stable platform for controlling knee valgus and foot pronation motions (33). However, given the static conditions during CA, their engagement was low. Therefore, this distinct effect of CA on mentioned muscles might somehow explain improved sprint time with a lack of impact on CMJ performance. Particularly, because the GM seems to be more involved during sprinting than the VL, while in the CMJ, it is the opposite; the VL seems more engaged than the GM (34,35). In this scenario, the CA affected the VL, which resulted in a lack of improvement during CMJ, whereas no change in the viscoelastic properties of the GM allows for enhancing sprint time. Nevertheless, the lack of changes in the viscoelastic properties of the GM muscle with changes in the VL muscle seems to confirm the strong evidence of a localized PAPE effect (24,44).

Recently, tensiomyography has been used to evaluate contractile properties due to CA-induced changes (6,22,37). However, as shown by the results of this and previous studies (related to fatigue), the measurement of acute changes in viscoelastic muscle properties may also be useful in examining the PAPE

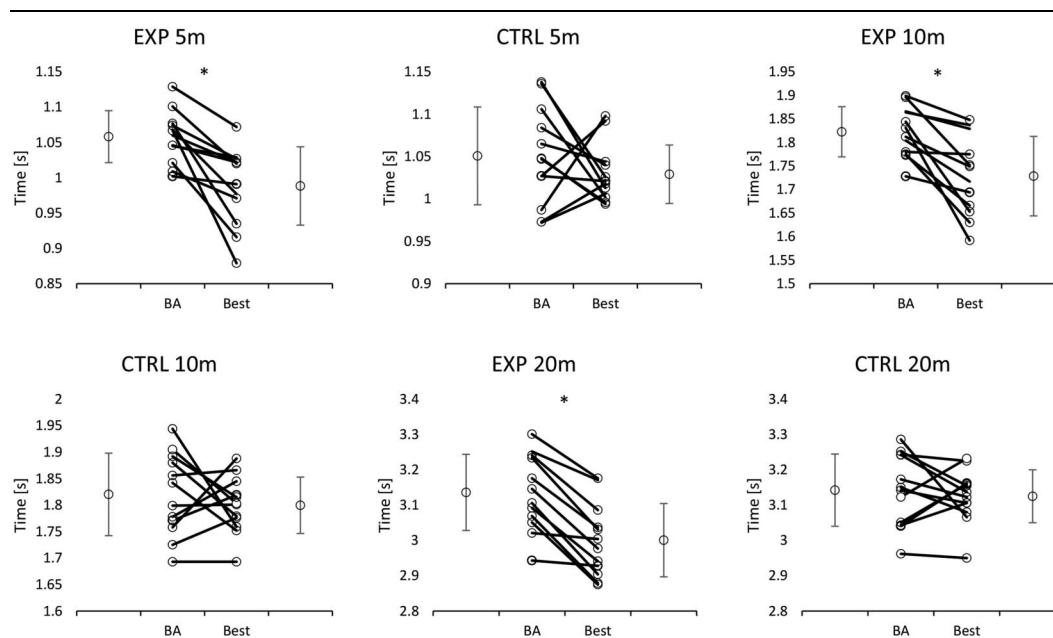


Figure 4. Mean $\pm$ SD and interindividual responses to the CA in sprint. CA = conditioning activity; BA = baseline; BEST = best performance postconditioning activity irrespective of the rest interval; EXP = experimental condition; CTRL = control condition.

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effect. Furthermore, the assessments using the Myoton device, as in the current study, seem to be much easier and faster. First, there is no need to place electrodes; second, there is no electric stimulation, so there is no need for rest intervals between measures. Therefore, it is possible to evaluate several muscle points simultaneously while measuring one with tensiomyography. Moreover, the device is smaller (similar to a smartphone) and thus easier to transport than tensiomyography. Although this study provides valuable information from a practical point of view, it should be interpreted cautiously. First, viscoelastic properties were analyzed only in the dominant limb; therefore, it is not known whether similar reactions also occurred in the

nondominant limb. Second, the subjects performed a CMJ without an arm swing, so it is not certain that the results of a CMJ with an arm swing, which is more natural for handball players, would have been different. Moreover, to the best of our knowledge, this is the first study that used the Myoton device and measured athletic performance to investigate the PAPE effect. Therefore, more research is needed to support the findings of this study and shed more light on this topic, especially considering previously reported differences in muscle stiffness between sexes (23). In addition, we suggest investigating other dynamic exercises acting as CAs and evaluation of the viscoelastic properties of other muscles and tendons.

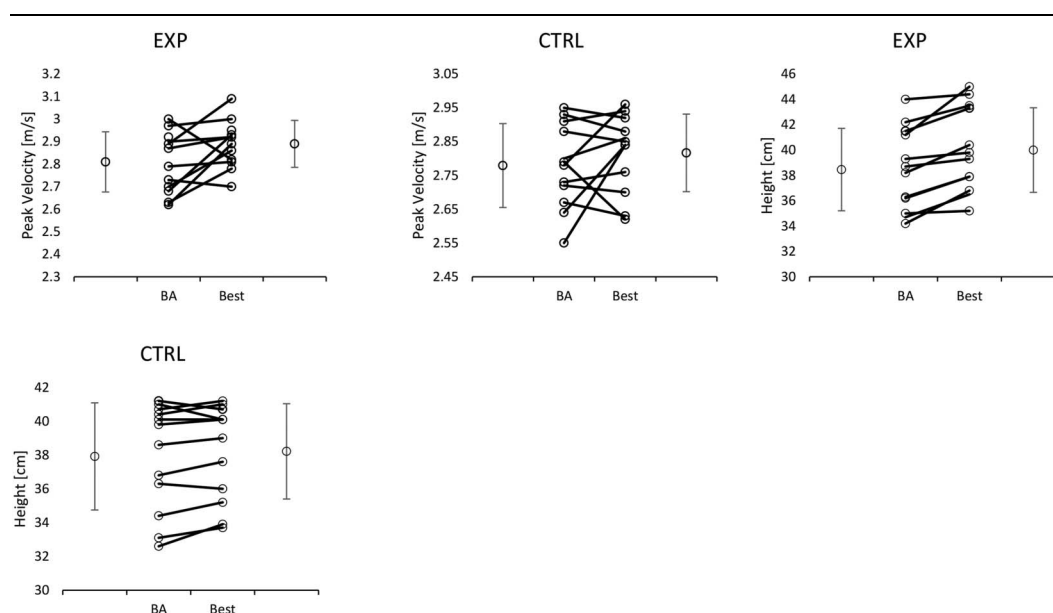


Figure 5. Mean $\pm$ SD and interindividual responses to the CA in countermovement jump. CA = conditioning activity; BA = baseline; BEST = best performance postconditioning activity irrespective of the rest interval; EXP = experimental condition; CTRL = control condition.

This study used an isometric CA, which has often been used in laboratory studies (3,4,17,18,31) and nonlaboratory studies (11) due to its high efficiency in inducing the PAPE effect. This was another study that confirmed the effectiveness of an isometric CA in improving subsequent athletic performance. The isometric protocol used by Bogdanis et al. (11) and replicated in this study resulted in a significant improvement ($ES =$ from 0.66 to 0.85) in sprinting performance within the next 4–12 minutes after isometric CA. In addition, in the case of jumping performance, a significant increase in jump height was noted. Although this effect was similar to that of the sprints (when the individual best time points after the CA were analyzed), this was not the case for particular time point analysis. This indicates a high interindividual variability for the optimal intra-complex rest interval among subjects and a shorter duration of the PAPE effect compared with sprinting. Therefore, the observed performance augmentation of CMJ only became apparent when the best attempts of the subjects were analyzed and not at all time points. The literature indicates that the PAPE effect is characterized by high interindividual variability, which may be related to the training experience, strength level, and the number of type IIx fibers in the exercised muscles (42,44). Furthermore, the data presented in the current study are partially in line with earlier findings that indicate that the PAPE effect is greater for sprinting than for jumping (42). The range of motion used during CA is a potential explanation for this finding. For example, Esformers and Bampouras (15) reported improvement in CMJ performance after both quarter and parallel squat exercises as the CA; however, the parallel back squat was superior. Moreover, Seitz and Haff (42) showed that shallow squat depths produced a considerably greater PAPE effect than deep squats. Therefore, it cannot be ruled out that if a smaller knee flexion angle had been used in the protocol of this study, the reported PAPE effect during the CMJ would have been greater. Nevertheless, the obtained results allow speculation that the PAPE “window of opportunity” seems much shorter for jumps than for sprints and shallow back squats might be superior to deep ones to potentiate jumping performance. Therefore, we suggest that future research concentrates on individual analysis of the PAPE effect among subjects and examines the influence of different ranges of motions during CAs. Moreover, the maximal isometric CA may be an attractive and safe solution to induce PAPE due to its easy practical application without requiring additional maximum strength testing and specific equipment. Although a significant decrease in muscle tone and stiffness during CA muscles may be evidence of performance enhancement, additional studies are needed to validate these findings further.

Practical Applications

Coaches and practitioners could use maximum isometric back squats as the CA during complex training sessions or as part of a precompetition warm-up to acutely improve sprinting and jumping performance. However, the individuals should not expect an increase in muscle stiffness after isometric CA. Nevertheless, to reach maximum benefit from a CA, including a maximal isometric strength protocol, individuals' responses, at which time the peak performance is reached, should be assessed. Our results suggest that the time to achieve the maximum PAPE effect in CMJ and sprinting might differ.

Acknowledgments

Blinded for peer review.

The authors report there are no competing interests to declare.

The research was supported by XXX under Grant XXX.

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